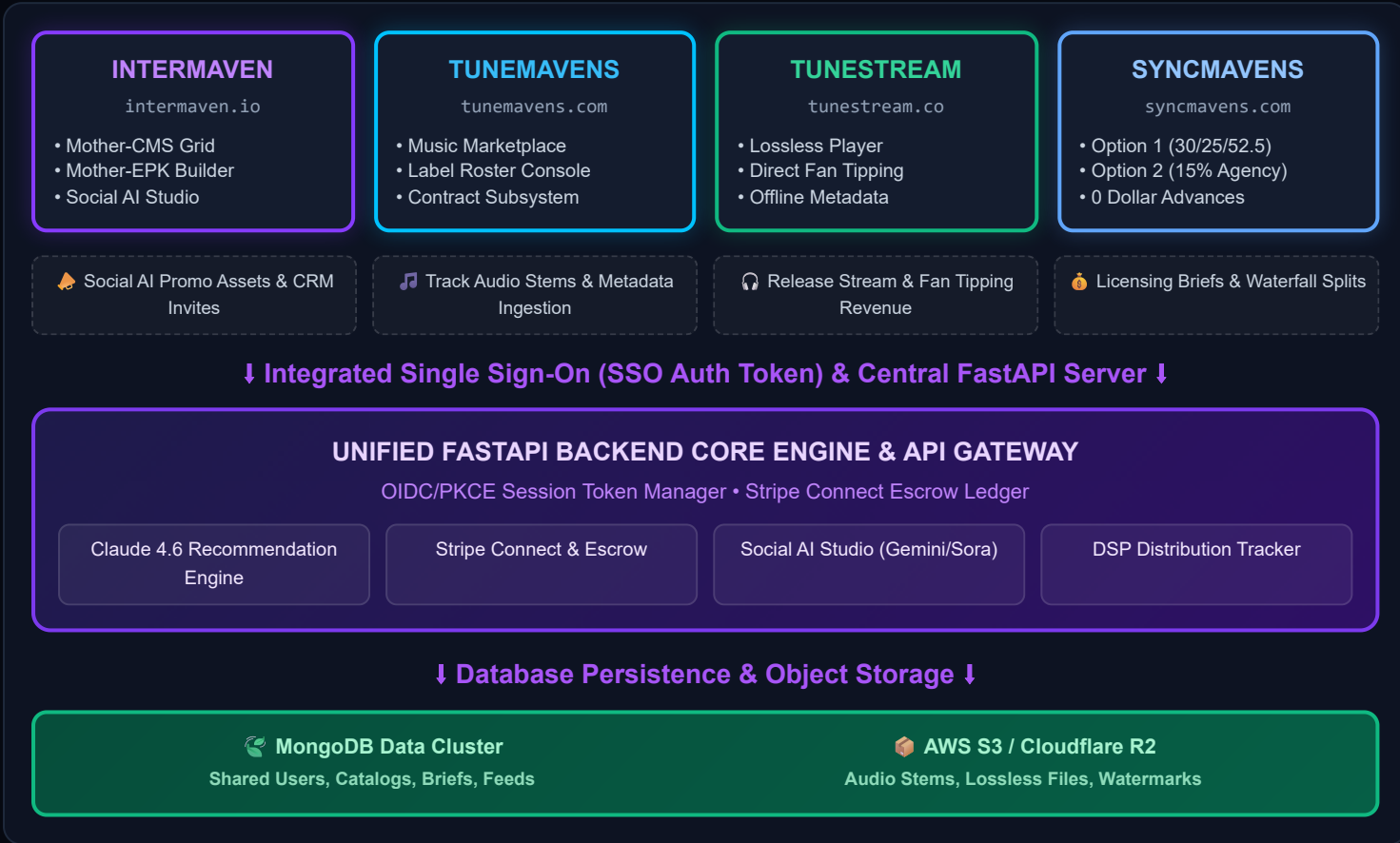


ECOSYSTEM SYSTEM ARCHITECTURE & FLOWCHART

Platform Interconnections, Multi-Directional Data Flow & Infrastructure Blueprint

1. Ecosystem Platform Connectivity & Inter-Application Data Flowchart



2. End-to-End Data & Financial Stream Lifecycle

- Catalog & Stem Ingestion:** Creator uploads audio stems to TuneMavens. Stems store in AWS S3, metadata upserts to MongoDB catalogs with auto-generated ISRCs.
- Split Cascade Lock:** Creator defines split waterfall (90/10) in publishing_deals and signs contract via contracts.
- Sync Placement Match & Revenue Payout:** Supervisor posts brief on SyncMavens. AI Simulator evaluates track compatibility. Pitch delivered. Supervisor approves and licensing waterfall executes (Option 1: 30% Agency / 25% Admin / 52.5% Creator Pool vs Option 2: 15% SyncMavens Agency Fee / 85% Creator Pool).
- Consumer Streaming & Fan Tipping:** Track publishes to TuneStream. Fans stream lossless audio & tip artist directly. Revenue credits route to unified network balance.
- Social AI Marketing:** Creator triggers Intermaven Social AI Studio to render 1:1 and 9:16 creative assets (Gemini/Sora) and dispatch via Resend email / social DM.

3. MongoDB Shared Collection Index

Collection Name	Key Fields & Attributes	Primary Consuming Apps	Business Purpose
users	_id, email, roles[], primary_role, plan, credits	All 4 Platforms	Unified SSO identity, role permissions, and credit balance
catalogs	_id, isrc, title, stems_s3_url, splits[]	TuneMavens, SyncMavens, TuneStream	Centralized metadata index for masters, stems, and audio assets
briefs	_id, supervisor_id, budget, deadline, genre	SyncMavens	Licensing opportunities submitted by music supervisors
pitches	_id, brief_id, catalog_id, match_score, waterfall_splits	SyncMavens, TuneMavens	Pitch submissions and licensing deal ledgers
publishing_deals	_id, user_id, tier, splits_cascade, status	TuneMavens	Publishing agreement records and split calculation rules
contracts	_id, contract_type, locked_clauses[], signatures[]	TuneMavens, SyncMavens	E-signature and legal contract negotiation state machine
campaigns	_id, name, segment, channels[], stats	Intermaven, TuneMavens	Multi-channel CRM growth marketing campaigns

4. Detailed Technical Stack Specifications & Monorepo Architecture

4.1 Monorepo Directory Layout & Application Boundaries

```
tunemaven/  
├── apps/  
│   ├── portal/           # Main Portal & Dashboard Hub (tunemavens.com)  
│   ├── tunestream/       # Consumer Audio Streaming App (tunestream.co)  
│   └── syncmavens/       # Sync Licensing Marketplace (syncmavens.com)  
├── packages/  
│   ├── shared-ui/        # Shared CSS Tokens, UI Components & Modals  
│   └── shared-auth/      # Shared OIDC/PKCE Session Helpers  
├── backend/  
│   ├── server.py         # Primary FastAPI Application Entry  
│   ├── auth.py           # PyJWT Cross-Domain Token Validation  
│   ├── routes/           # Domain Routers (sso, match, stream, crm, cms)  
│   └── models.py         # Pydantic v2 Data Transfer Models  
└── deploy/              # Multi-Domain Hostinger VPS Nginx Configs
```

4.2 Layer-by-Layer Technical Specification Matrix

Tech Layer	Platform Standard	Implementation & Package Specification
Frontend Framework	React 18 + Vite	Modular Monorepo Monolith with Isolated Build Targets (dist/portal, dist/tunestream, dist/syncmavens).
Styling System	Vanilla CSS + Tokens	Custom HSL design tokens, glassmorphism, responsive grid layouts, Outfit/Inter fonts. Zero CSS framework bloat.
Native Packaging	Capacitor	Wraps web frontend applications for iOS & Android native deployment with native audio drivers.
Backend Engine	FastAPI (Python 3.11)	ASGI asynchronous web server running Uvicorn workers with non-blocking event loops.
Database Layer	MongoDB + Motor	Multi-tenant MongoDB replica set cluster using Motor for async Python database I/O.
Object Storage	AWS S3 / Cloudflare R2	Direct browser presigned URL upload pipeline for lossless audio stems (WAV/AIFF) and preview clips.
Single Sign-On	OIDC + PKCE	Custom OAuth2/OIDC SSO server (sso_router.py) issuing PyJWT cookies across .tunemavens.com.
Payment Gateway	Stripe Connect	Direct payouts, 4-tier entitlements, QR event ticketing scanner, Option 1 (30/25/52.5) & Option 2 (15% agency fee) ledgers.

Tech Layer	Platform Standard	Implementation & Package Specification
AI Recommendation	Claude Sonnet 4.6	15s capped recommendation synthesis engine (<code>users_router.py</code>) recommending tools and apps.
Social AI Studio	Gemini Nano + Sora 2	Multi-format image (<code>/generate-art</code>) and short-form video clip (<code>/generate-video</code>) campaign generator.
CRM Engine	Resend API + Inbox	Multi-channel messaging engine dispatching emails via Resend and in-app inbox threads (<code>messages</code>).